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📅 14/05/2004

🛡 Military Status: Exempted

🐙 GitHub

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📍 Giza, Egypt

🌐 LinkedIn

🔗 Portfolio

Profile

Backend and DevOps engineer completing a BSc in Computer Science at KSIU (June 2026). Delivered four R&D prototypes adopted as production handoff artifacts by NBK Egypt and built ACR-QA, an AI-powered code review platform that outperforms enterprise tools. Engineered an observability stack sustaining 225 req/sec at 62ms P95 latency under Egypt's nationally sponsored DEPI program. Skilled in Rust, C++, Python, and Java across AI systems, banking infrastructure, and containerized deployments. Available for remote and on-site roles from July 2026.

Professional Experience

DevOps Engineering Trainee 06/2025 – 12/2025 | Zamalek, Giza, Egypt

Digital Egypt Pioneers Initiative (MCIT – EYouth).

- Selected for a national DevOps training program sponsored by MCIT.
- Automated CI/CD pipelines with Git, Jenkins, and Docker across a **5-stage GitHub Actions workflow**
- Built reusable IaC modules with **Terraform** and **Ansible** to provision **5+ environments**.
- Deployed, monitored, and optimized containerized microservices on **Kubernetes** clusters using **Prometheus** and **Grafana** for observability
- Capstone: production URL shortener with **225 req/sec throughput**, **99.93% uptime**, **P95 latency 62ms** — 19 Grafana panels across 3 dashboards, Kubernetes HPA, Terraform IaC, 5-stage CI/CD.

R&D Intern — Development & Application 08/2025 – 09/2025 | New Cairo, Cairo, Egypt

National Bank of Kuwait - Egypt

- Built and delivered AI/ML prototypes for NBK Egypt's R&D division — including a **loan risk predictor** (Streamlit, RandomForest, SMOTE) and an **OFAC sanctions screening system** (Python, TF-IDF, fuzzy logic).
- Developed a **secure PDF retrieval service** (Spring Boot, RSA/AES) integrated with **Temenos T24** via Tomcat.
- Built **Outlook→SFTP automation** (Java, JACOB COM bridge, JSch) to extract and securely transfer email attachments to Linux servers, replacing a manual banking workflow.
- Applied SOLID principles, design patterns, and Docker containerization across all **4 prototypes**.
- Internship extended by **2 weeks based on performance** — all 4 prototypes delivered as **production handoff artifacts** with RUNBOOK documentation.

Information Technology Intern 08/2024 | Sheikh Zayed, Giza, Egypt

Emkay Foods

- Supported IT operations across **150+ endpoints**: Active Directory, DHCP, VPN/VLAN configuration, and managed switch/firewall setup (**FortiGate**, **Sophos**)
- Assisted with ERP user provisioning and access management, improving onboarding and ticket turnaround

Projects

ACR-QA — Automated Code Review Platform 12/2025 – 05/2026

Graduation Project · Python · RAG · Docker · PostgreSQL · GitHub

Actions

- Built a language-agnostic static analysis platform running **7 tools in parallel** (Ruff, Semgrep, Bandit, Vulture, Radon + custom secrets/SCA scanners) with a canonical normalization schema across all tools
- Implemented **RAG-enhanced AI explanations** grounded in a **66-rule knowledge base**; added semantic entropy scoring (**3× LLM runs** for consistency) and hallucination detection at **\$0 recurring cost**
- Delivered a **273-test** pytest suite passing in **under 6 seconds**; GitHub Actions + GitLab CI integration; SARIF v2.1.0 export; OWASP Top 10 compliance mapping with CWE IDs
- Built custom **AST-based Test Gap Analyzer** and **Policy-as-Code engine** with schema validation — features absent from CodeRabbit and SonarQube Enterprise

Redis from Scratch (Rust, TCP, Multithreading, Concurrency, Systems Programming) 03/2025 – 06/2025

Actions

- Built a **multithreaded in-memory key-value store** with support for Strings, Lists, Sets, Sorted Sets, and TTL.
- Implemented the command parsing layer supporting **16 Redis commands** across **4 data structures** including TTL-based key expiry and LRange/ZRange with index bounds.
- Ensured concurrency safety and atomic operations using **Arc**, **Mutex**, and Rust's ownership model.

Containerized URL Shortener — Production Monitoring Stack 

DEPI Graduation Project · Node.js · Docker · Kubernetes · Prometheus · Grafana · GitHub Actions · Terraform

- Built and containerized a URL shortener REST API (Node.js + Express + SQLite) with custom Prometheus metrics instrumentation — **19 Grafana panels** across **3 dashboards** (main monitoring, analytics, system health)
- Achieved **225 req/sec throughput**, **99.93% uptime**, and **P95 latency of 62ms** under k6 load testing with **75 concurrent users**
- Implemented full DevOps lifecycle: **5-stage GitHub Actions CI/CD**, Kubernetes deployment with **HPA** and PodDisruptionBudgets, **Terraform IaC** modules, and multi-channel alerting (Slack, Email, Discord)
- Added automated disaster recovery with **15-minute RTO**, Docker volumes for persistence, and a security audit (**B+ grade**, zero critical vulnerabilities)

10/2025 – 11/2025

DecodedSound — AI Cultural Translation Engine 

Next.js 15 · TypeScript · Groq LLaMA · Whisper · PostgreSQL · Prisma · Docker

- Engineered a **3-input pipeline** (typed lyrics, audio upload, YouTube URL via yt-dlp) → Groq Whisper transcription → **LLaMA 3.3 70B** translation → **4-panel output** (plain translation, line-by-line annotation, slang dictionary, cultural context).
- Built a full-stack AI translation platform for SDK (Esdeekid) music — a hyper-coded Cape Flats dialect mixing Kaaps Afrikaans, Xhosa, and prison gang codes that no existing AI model handles correctly.
- Built a living knowledge base of **502 hand-curated slang terms** with community contributions, AI candidate discovery, admin moderation panel, and auto-downgrade on **3 community flags**.
- Implemented **12 REST API routes**, IP-based **sliding-window rate limiting** across 3 tiers, YouTube caching (translated URLs served instantly on repeat).

02/2026 – 05/2026

Education

BSc in Computer Science – Software Engineering

King Salman International University

- Activities: IEEE, GDSC, ICPC communities, COP27 Youth Volunteer, Huawei ICT Academy
- Achievement: ICPC ECPC 2024 — Honorable Mention, 2nd place among KSIU teams

10/2022 – 06/2026 | South Sinai, Egypt

Skills

- Languages:** Rust, C++, Python, Java, TypeScript, JavaScript, HTML, CSS
- Backend:** Node.js, Express, FastAPI, Flask, Spring Boot, REST APIs, PostgreSQL, Redis, MongoDB, SQLite, Prisma, JWT, OAuth
- DevOps:** Docker, Docker Compose, Kubernetes, Terraform, Ansible, Jenkins, GitHub Actions, GitLab CI, Prometheus, Grafana, Nginx
- ML & AI:** scikit-learn, RAG, Ollama, Groq, LLaMA, Whisper, SMOTE, PCA, RandomForest, Pandas, NumPy, Streamlit
- Infra & Security:** Linux, AWS, GCP, SSH, systemd, SFTP, RSA/AES encryption, TCP/IP, DNS, DHCP, VLANs, VPNs, NAT, FortiGate, Sophos, Active Directory
- Testing & Observability:** pytest, k6, Apache Bench, SARIF, OWASP
- Tools & Practices:** Git, OOP, data structures, concurrency, multithreading, microservices, SOLID principles, clean architecture

Languages

Arabic

Mother Tongue

English

Fluent